

Notes: Rogoff & You (RES, 2023)

Redeemable Platform Currencies

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1 Big picture

- **Question.** Can large online platforms issue redeemable digital tokens that function partly like platform money, and should platforms make those tokens tradable?
- **Setting.** The tokens are redeemable for platform goods or services but not necessarily for cash. They resemble gift cards, stored-value balances, loyalty points, in-game currencies, and platform-specific digital cash.
- **Main friction.** The platform has a higher return on outside funds than consumers have on retail bank savings. Token issuance can therefore move resources forward to the platform and create gains from intertemporal trade.
- **Central result.** In the baseline one-time ICO, non-tradable tokens can earn higher issuer revenue than tradable tokens of the same supply.
- **Core intuition.** Tradability helps a consumer who wants to sell tokens she already has, but it also lets future consumers buy tokens from other consumers instead of holding large balances upfront. This reduces the platform’s ability to charge high prices in the initial sale.
- **Economic takeaway.** Platform token design is not just a technological issue. Tradability changes market power, timing of cash flows, consumer surplus, and the feasibility of intertemporal price discrimination.

2 Incremental contribution of the paper

- **Redeemability, not cryptocurrency alone.** The paper shifts attention from decentralized crypto assets to redeemable claims issued by platforms and backed by future in-platform goods.
- **Non-tradability as a strategic feature.** Much token research emphasizes liquidity and tradability. This paper shows that non-tradability can raise issuer revenue because it preserves monopoly power over future token demand.
- **Bridge between loyalty programs and digital currencies.** The paper connects Green Stamps, frequent-flyer points, stored-value cards, and token-based platforms in one framework.
- **Simple mechanism for ICO pricing.** The baseline model gives closed-form token prices and revenues, making clear that tradability compresses redemption timing and lowers the ICO price.
- **Policy relevance.** The model clarifies when a platform token is mainly a loyalty/financing instrument and when it could become a general-purpose money-like asset.

3 Data and measurement / model primitives

3.1 Historical taxonomy of redeemable platform assets

The paper organizes redeemable assets into four generations: physical trading stamps, centralized loyalty points, centralized platform cash/stored-value balances, and cryptographic/blockchain/cybersecurity tokens.

Interpretation: The historical evidence motivates the model. Restrictions on tradability are not incidental; they are a recurring design choice in redeemable platform assets.

3.2 Consumers, platform purchases, and tokens

Time is discrete. In each period a consumer receives an exogenous platform-consumption shock with probability p . One token can be redeemed for one unit of platform consumption, with fiat value normalized to

one. Consumers can hold wealth outside the platform and earn retail return r , with $\beta(1+r) = 1$.

$$W_{t+1} = (1+r)(W_t - \Psi_t - \Theta_t).$$

Interpretation: The consumer compares token purchases with holding retail savings. Token demand is based on future use value, not intrinsic cash redemption.

3.3 Platform discounting and gains from token issuance

The platform has discount factor $\beta^* < \beta$. This wedge is the only source of gains from intertemporal trade in the baseline model. Token issuance lets the platform receive cash today while delivering goods later.

Interpretation: The platform is not issuing tokens because tokens reduce production costs. It issues tokens because it values early cash more than consumers do.

4 Models and theoretical specifications

4.1 Non-tradable ICO

For a one-time ICO of M non-tradable tokens, the marginal price is:

$$P_{I,N}(M) = \left(\frac{\beta p}{1 - \beta(1-p)} \right)^M.$$

The issuer's revenue includes upfront token revenue and later fiat revenue after tokens are depleted.

Interpretation: A larger ICO raises more token revenue but lowers the price on all tokens and delays future fiat sales.

4.2 Tradable ICO

When all consumers hold at most one token, the steady-state token price satisfies:

$$P = \beta p + \beta(1-p)P \quad \Rightarrow \quad P = \frac{\beta p}{1 - \beta(1-p)}.$$

For M tradable tokens:

$$P_{I,T} = \beta^{(M-1)/p} \left(\frac{\beta p}{1 - \beta(1-p)} \right).$$

Interpretation: Tradability creates a resale market. This lets consumers avoid holding large inventories, lowering initial willingness to pay for a large ICO.

4.3 Non-tradability revenue dominance

Lemma 1 states:

$$\beta^{1/p} < \frac{\beta p}{1 - \beta(1-p)}.$$

Proposition 1 states:

$$R_{I,N} > R_{I,T}.$$

Interpretation: Non-tradability dominates because tradable tokens are spent faster and push platform fiat revenue further into the future.

4.4 Optimal issuance quantity

The platform chooses M to maximize token revenue minus the discounted value of foregone fiat-money revenue:

$$\max_M \left\{ M \left(\frac{\beta p}{1 - \beta(1 - p)} \right)^M - \sum_{i=1}^M \left(\frac{\beta^* p}{1 - \beta^*(1 - p)} \right)^i \right\}.$$

Interpretation: Optimal issuance is a nonlinear pricing problem. Issuing more tokens changes the price of every token, not just the marginal token.

4.5 Price menu, SCO, and heterogeneity

A non-tradable price menu lets the platform charge each token according to marginal value and can extract all gains from trade. In contrast, tradability destroys the menu through resale arbitrage. With memory-less SCOs, Proposition 2 states that any constant-supply equilibrium has $M = 1$ regardless of tradability. Proposition 4 shows that the main non-tradability result survives consumer heterogeneity.

5 Identification and theoretical design

This is a theoretical paper. The relevant identification question is how the model isolates the mechanism behind token design. The baseline comparison holds fixed the token face value, redemption value, consumption probability, and platform technology, then varies only the tradability rule. This identifies the role of the resale option.

- **Commitment.** The platform can commit to token quantity and issuance policy.
- **Redeemability.** Tokens are redeemable for platform goods but not necessarily for cash.
- **No outside convenience yield.** Baseline tokens have no value outside the platform.
- **No broad network adoption.** The platform token does not become general-purpose money in the benchmark.
- **Exogenous platform consumption.** Consumption shocks arrive with probability p .

Interpretation: The mechanism is identified by a clean counterfactual: same token, same platform, same consumers, different tradability rule.

6 Tables and figures

6.1 Table 1: Generations of redeemable platform tokens

Read: The table compares physical trading stamps, centralized loyalty points, centralized platform cash, and cryptographic/blockchain/cybersecurity tokens.

Detailed interpretation: The table shows that restrictions on transferability are historically common and economically meaningful.

How it fits the paper: It motivates the main model by showing that real-world platform assets combine redeemability with controlled liquidity.

TABLE 1
Generations of redeemable platform tokens

Generation	Technology	Circulation mechanism	Leading examples	Redeemability	Tradability	Size
First Generation: TRADING STAMPS	Physical stamps (anonymous)	Sold wholesale to retailers then pro-rata to consumers	S&H Green stamps (US), Green Shield stamps (UK)	Convertible to awards at trading stamp stores, or by mail, after filling savers' books	Often legally prohibited, but black market	Peaked in the mid-60s when S&H was printed 32 million catalogues and 140 million savers books
Second Generation: LOYALTY POINTS	Centralized Accounting	Bundled with sales, bonus mechanisms for frequent buyers	Airline and hotel loyalty points, also supermarket and drug stores	Convertible for services of issuing company or partners	Typically constrained with limited black markets	American Airline awards 7-8% of yearly revenue miles for points (8.5 billion outstanding in 2018). Stock equals 20% one year's gross revenue
Third Generation: PLATFORM CASH	Centralized Accounting	Often sold at discount. Very convenient for in-platform use	Uber & Lyft cash, Starbucks stored-value cards, Amazon gift cards, Q-Coins, Gash+	Generally redeemable only on platform but still evolving	Generally none or highly constrained	Uber cash, Starbucks balance, 3.3 billion dollar Amazon gift cards unredeemed in 2019
Fourth Generation: TOKENS ^a	Cryptography, Blockchain, Cybersecurity Technologies	Initial and recurrent coin offerings (evolving), aims to out-platform uses even to the places with little financial arms	Ethereum, Telegram, Facebook Diem (2021), Alipay, Wechat Pay	Can be redeemable for platform services, but usually aspires to universal usage	Tradable or tradable with minor restrictions	Ether can be used for smart contract services, Ether topped over \$500 billion in May, 2021

Stable coins are a special case also redeemable for cash.

Table 1: Generations of redeemable platform tokens. The table classifies each generation by technology, circulation mechanism, examples, redeemability, tradability, and scale.

6.2 Figure 1: Utility convexity; Figure 2: Probability distribution of the last token spent

Read: Figure 1 shows the convex discounting logic. Figure 2 compares last-token-spending distributions for non-tradable and tradable ICO tokens.

Detailed interpretation: Tradability compresses redemption timing and lowers the initial price; non-tradability creates slower, more dispersed redemption, which raises upfront demand.

How it fits the paper: These figures are the visual intuition for Lemma 1 and Proposition 1.

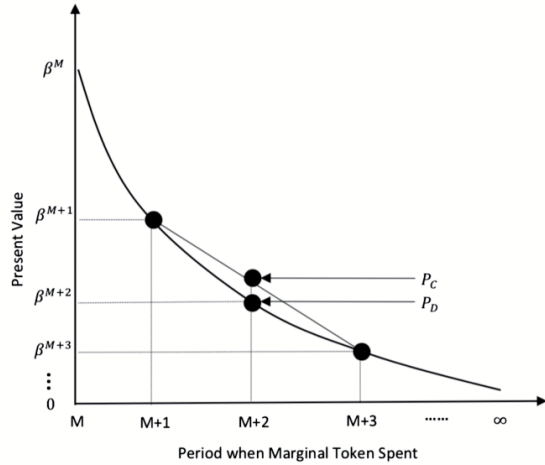


FIGURE 1
Utility convexity

Figure plots the convex β -discounting function and shows the prices of the following two lotteries: Lottery 1 (price P_C) and Lottery 2 (price P_D): One dollar in period $M+1$ with 50% probability, and one dollar in period $M+3$ with 50% probability. Convexity implies $P_C > P_D$.

(a) Figure 1: Utility convexity.

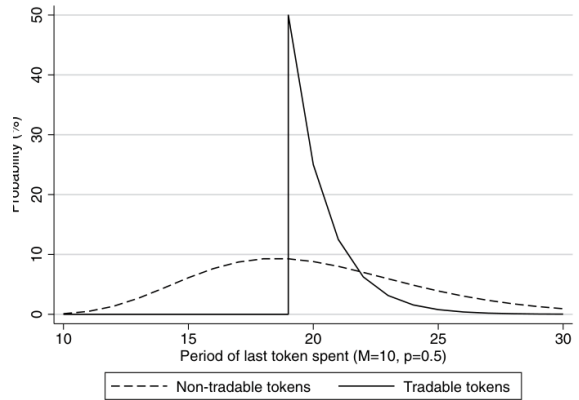


FIGURE 2
Probability distribution of the last token spent

Figure plots the probability the period of the last token spent for tradable and non-tradable ICO respectively. We pick M and the probability of consumption is $p=0.5$. The solid line plots the probability distribution function of tradable tokens and the dashed line plots the probability distribution function of non-tradable tokens.

(b) Figure 2: Probability distribution of the last token spent.

6.3 Figure 3: Optimal issuance quantity from three to two; Figure 4: Optimal issuance quantity from three to four

Read: Figure 3 shows the tradeoff from reducing issuance; Figure 4 shows the tradeoff from increasing issuance.

Detailed interpretation: Issuing more tokens raises quantity sold but lowers the price on all tokens and delays fiat revenue.

How it fits the paper: The figures are the graphical version of the optimality inequalities for M .

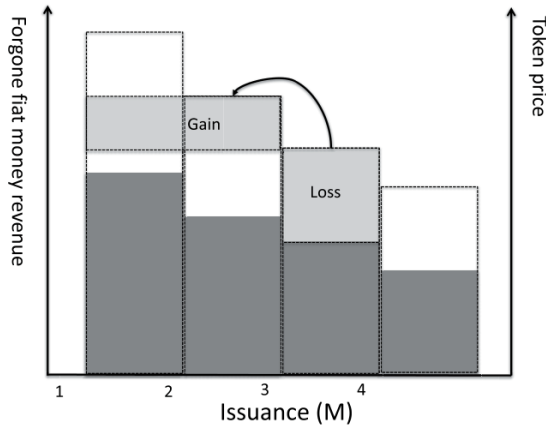


FIGURE 3
Optimal issuance quantity: from three to two

Figure shows the gain and loss by reducing one token (from three to two). The issuer gains from the price increase, but

(a) Figure 3: Optimal issuance quantity: from three to two.

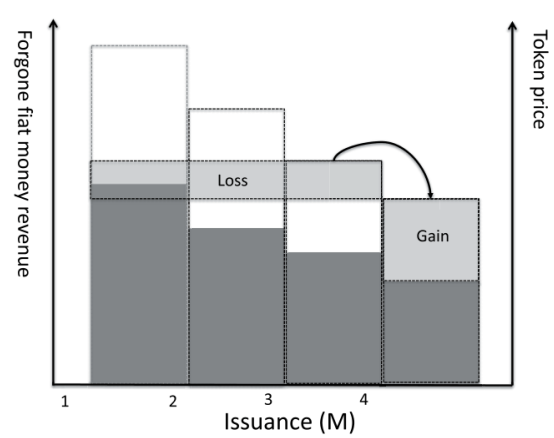


FIGURE 4
Optimal issuance quantity: from three to four

(b) Figure 4: Optimal issuance quantity: from three to four.

6.4 Figure A.1: Effective discount factor

Read: Figure A.1 plots the difference between effective discount factors for non-tradable and tradable tokens as a function of β .

Detailed interpretation: A positive curve means the non-tradable effective discount factor is higher, so adding token supply reduces price more slowly.

How it fits the paper: The figure supports the effective-discount-factor inequality underlying the main revenue ranking.

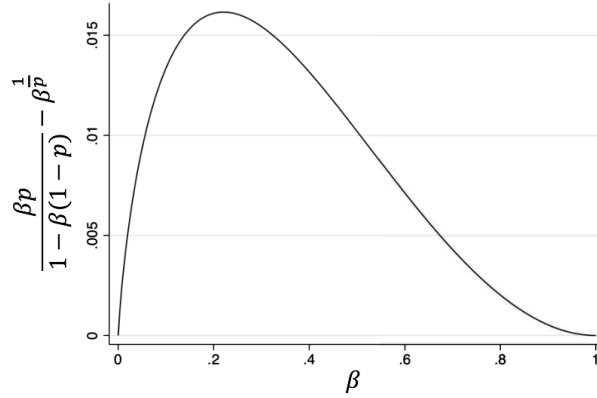


FIGURE A.1
Effective discount factor

test: This figure plots the difference between effective discount factors for the non-tradable and tradable tokens as a function of β . The stability of consumption shock is set as $n=0.9$.

Figure A.1: Effective discount factor.

6.5 Appendix Figures B.1–B.4: proof checks for heterogeneity

Read: The appendix figures visualize monotonicity and positivity properties used in the proof with heterogeneous consumption probabilities.

Detailed interpretation: They show that the non-tradability result does not rely only on homogeneous consumers.

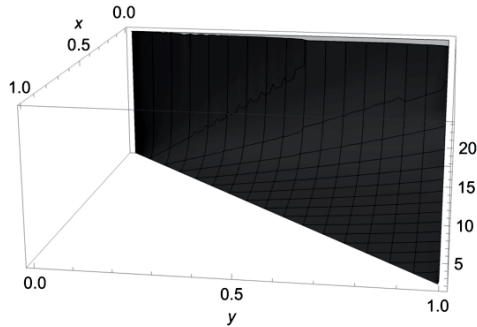


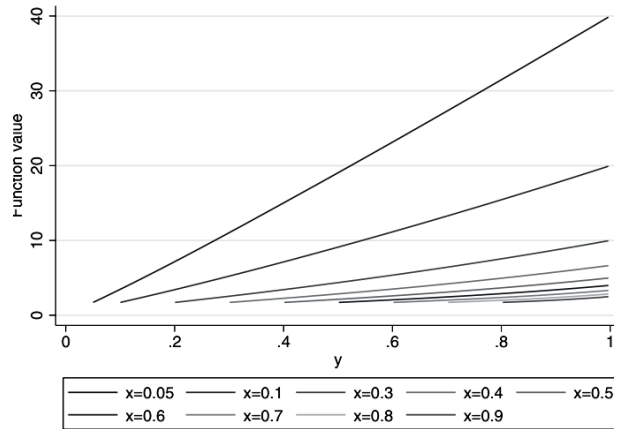
FIGURE B.1

Three-dimensional surface of RHS monotonicity

figure plots the three-dimensional surface of the RHS as a function of (x,y) .

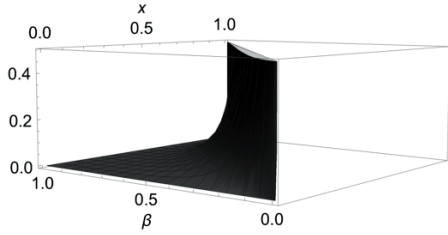
... of Figure 2: ICC under homogeneity with heterogeneous...

(a) Figure B.1: RHS monotonicity surface.

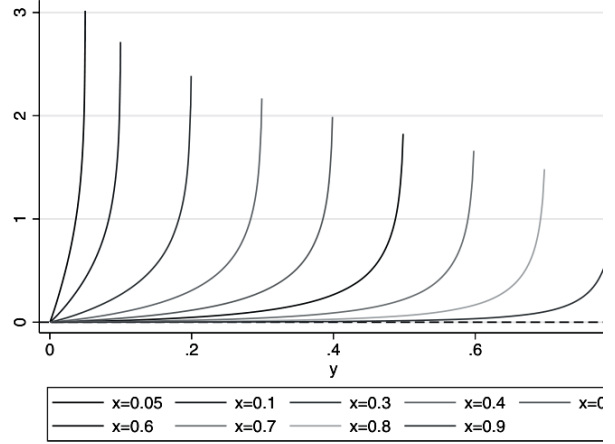


(b) Figure B.2: RHS monotonicity curve.

FIGURE B.3
Two-dimensional curve of RHS monotonicity
figure plots the RHS as a function of y , given $x=0.05, 0.1, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8$, and 0.9 .



(a) Figure B.3: $G(\beta, x)$ surface.



(b) Figure B.4: $G(\beta, x)$ curve.

7 Short summary

- The paper studies redeemable platform tokens that can be used for platform purchases but are not necessarily redeemable for cash.
- The platform issues tokens because it values early cash more highly than consumers value money held outside the platform.
- In a one-time ICO with homogeneous consumers, non-tradable tokens earn higher issuer revenue than tradable tokens for any fixed issuance size.
- Tradability lowers the initial token price because consumers know they can buy tokens later in the resale market.
- Non-tradability also makes price menus and memory-based issuance more effective.
- Tradability can be optimal only if it creates enough outside-platform convenience yield, general payment value, or network benefits.

8 Conclusion

8.1 Core conclusion

Rogoff and You show that platform token tradability is not automatically desirable from the issuer's perspective. In the benchmark model, non-tradable redeemable tokens dominate tradable tokens because tradability creates a resale market that competes with the platform's own future token demand and weakens the platform's ability to front-load revenue.

8.2 Economic intuition

A consumer likes the ability to sell tokens she already owns, but she also anticipates being able to buy tokens later from others. That expectation lowers willingness to purchase a large token balance in the initial offering. Non-tradability forces each consumer to self-insure against future platform purchases by holding tokens, which raises upfront demand and issuer revenue.

8.3 Platform-design implications

Non-tradability supports price menus, bulk discounts, and memory-based issuance. Tradability becomes attractive only if it adds a new service: broad payment convenience, outside-platform liquidity, or network adoption.

8.4 Regulatory implications

A closed-loop non-tradable platform token is closer to a loyalty or stored-value system. A widely tradable token can raise monetary, payments, consumer-protection, and financial-stability questions. Regulators should recognize that non-tradability can be a profit-maximizing restriction rather than a mere technological constraint.

8.5 Limitations and open questions

The model is partial equilibrium and does not estimate empirical magnitudes. It abstracts from platform competition, risk aversion, regulation, and broad macro-financial effects. Future work could quantify convenience yields, compare platforms with different resale restrictions, and study competition among platform currencies.

8.6 Takeaway

Platform currencies are not merely payment technologies. They are instruments of intertemporal pricing, liquidity design, and platform market power. Tradability improves user flexibility, but non-tradability can be the issuer's optimal design when the token's main value is in-platform redemption rather than broad monetary use.